Quiz 4

Week 6: forecasts, non-representative polls, uncertainty, measurement and sampling, and ISLR/P Chapters 4.3 and 5.1

Rohan Alexander

21 October, 2026

Name: _____ Student number: _____

Instructions

Part A: Multiple choice (questions 1 to 38). Please answer on the answer sheet below, by filling in one box completely for each question. Each question has one best answer. If a question has a blank, choose the option that best completes the sentence. Only the answer sheet is marked for Part A.

Part B: Short answer (questions B1 to B5). Please write neatly in the space provided, a few sentences each.

Answer sheet

1	a	b	c	d	**16**	a	b	c	d	**31**	a	b	c	d
2	a	b	c	d	**17**	a	b	c	d	**32**	a	b	c	d
3	a	b	c	d	**18**	a	b	c	d	**33**	a	b	c	d
4	a	b	c	d	**19**	a	b	c	d	**34**	a	b	c	d
5	a	b	c	d	**20**	a	b	c	d	**35**	a	b	c	d
6	a	b	c	d	**21**	a	b	c	d	**36**	a	b	c	d
7	a	b	c	d	**22**	a	b	c	d	**37**	a	b	c	d
8	a	b	c	d	**23**	a	b	c	d	**38**	a	b	c	d
9	a	b	c	d	**24**	a	b	c	d					
10	a	b	c	d	**25**	a	b	c	d					
11	a	b	c	d	**26**	a	b	c	d					
12	a	b	c	d	**27**	a	b	c	d					
13	a	b	c	d	**28**	a	b	c	d					
14	a	b	c	d	**29**	a	b	c	d					
15	a	b	c	d	**30**	a	b	c	d					

Part A: Multiple choice

The readings

1. From Gelman et al. (2020), a good presidential election forecast combines _____

a. fundamentals, polls, and a model of how opinion changes over time
b. polls alone, each weighted by its sample size and how recent it is
c. betting markets, adjusted for the bias towards the favourite
d. fundamentals alone, since polls are too noisy until the final week

2. From Gelman et al. (2020), why is it hard to check whether election forecasts are calibrated?

a. Official results are revised for months, so forecasts cannot be scored promptly.
b. Elections are few, and state outcomes are correlated rather than independent tests.
c. Forecasters publish expected vote shares and do not publish win probabilities.
d. A probability cannot be compared with a yes or no outcome, even in principle.

3. From Gelman et al. (2020), which incentive might push a forecaster to report more uncertainty than the evidence supports?

a. A confident forecast attracts more readers, which raises the forecaster's profile.
b. A wide interval scores better than a narrow one under every scoring rule.
c. A forecast that moves from day to day gives journalists more to write about.
d. A hedged forecast is rarely called wrong, which protects a reputation.

4. From Gelman et al. (2020), a forecast that gives a candidate a 70 per cent chance of winning means that _____

a. about 70 per cent of recent polls show the candidate ahead
b. in situations like this one, the candidate would lose about three times in ten
c. the forecaster is 70 per cent sure that this year's polls are accurate
d. the candidate is expected to receive about 70 per cent of the votes cast

5. From Wang et al. (2015), the sample was _____

a. Xbox users who answered daily polls before the 2012 US election
b. subscribers who returned a mail-in ballot to the Literary Digest in 1936
c. Amazon Mechanical Turk workers recruited in the final month of 2012
d. a random-digit-dial telephone sample fielded before the 2012 election

6. From Wang et al. (2015), in what way was that sample unrepresentative?

a. Most respondents lived in a small number of swing states.
b. Most respondents were not registered to vote in 2012
c. 93 per cent were men, and 65 per cent were aged 18 to 29
d. 93 per cent were over 65, and most lived in rural areas.

7. From Wang et al. (2015), what is the lesson of the 1936 Literary Digest poll?

a. Readers of magazines change their minds more often than other voters do.
b. A mail survey is less accurate than a survey conducted by telephone.
c. Two million responses is still too few to forecast a national election.
d. A very large sample does not help if it is biased towards one kind of person.

8. From Wang et al. (2015), the two steps of multilevel regression and poststratification are _____

a. average the available polls, then correct by the polling error at the previous election
b. drop over-represented respondents until the sample matches the census, then take the mean
c. model the outcome within demographic cells, then weight the cell estimates by population share
d. weight each respondent by the inverse of their response probability, then take the mean

9. From Wang et al. (2015), why use a multilevel model for the first step rather than the raw mean in each cell?

a. A multilevel model removes the need for population counts in each cell.
b. A multilevel model can be fitted without any demographic information.
c. Partial pooling gives stable estimates for cells with few respondents.
d. A multilevel model guarantees that the estimate for each cell is unbiased.

10. From Wang et al. (2015), what does the poststratification step need that the survey itself cannot supply?

a. The voter file for each state, matched to the individual respondents.
b. The national total for each demographic variable taken one at a time.
c. A second, independent survey of the same respondents, for validation.
d. Population counts for every cell, which the paper took from exit polls.

11. From Wang et al. (2015), how does the Xbox paper relate to your silicon sample?

a. A silicon sample is also non-representative, so it needs adjusting too.
b. A silicon sample has no sampling error, so no adjustment is needed.
c. A silicon sample is representative by construction, since its personas follow the census.
d. A silicon sample can be adjusted only by changing the prompt and running it again.

12. From Wang et al. (2015), why were party identification and 2008 vote included in the cells when the census does not record them?

a. They were used only to remove the respondents who said they had not voted in 2008
b. The Xbox platform required them from every respondent before a poll could be answered.
c. They explain much of the gap between the sample and the electorate, so exit polls supplied the targets.
d. The census does record them, in the long-form questionnaire.

13. From Wang et al. (2015), how were the adjusted estimates checked?

a. By calling a sample of the same respondents on the telephone and asking the question again.
b. By comparing them day by day with averages of conventional polls, and then with the result.
c. By running the same daily poll on a second gaming platform with a different audience.
d. They were not checked, because poststratification removes bias by construction.

14. From Kay et al. (2016), a quantile dotplot is _____

a. a scatterplot of past forecasts and the outcomes that followed them
b. a density plot with its quartiles marked as vertical lines
c. a histogram of the raw data, drawn with one dot for each observation
d. a predictive distribution drawn as a few dots, each one a quantile

15. From Kay et al. (2016), how does a reader get a probability from a quantile dotplot?

a. Read off the height of the curve at the threshold, as a percentage.
b. Count the dots on one side of the threshold and divide by the total.
c. Compare the position of the mean with the position of the threshold.
d. Judge by eye the share of the area that lies beyond the threshold.

16. From Kay et al. (2016), how were the displays of uncertainty compared?

a. People answered the same bus-arrival questions from each display, and the spread of their answers was compared.
b. A panel of designers rated each display for how attractive it looked on a phone screen.
c. The displays were compared on how much screen space each one took up on a phone.
d. Bus riders were asked which display they would prefer, with no questions about probability.

The lecture

17. From the Week 6 lecture, a valid measurement is one that _____

a. has been collected by a government statistical agency rather than a private one
b. has no missing values and no values outside the expected range
c. gives the same result when the measurement is repeated
d. relates directly to the estimand and the research question

18. From the Week 6 lecture, a reliable measurement is one that _____

a. was taken with an instrument that has been calibrated recently
b. is recorded in the units that the reader of the paper expects
c. gives similar results when the same thing is measured again
d. relates directly to the research question and the estimand

19. From the Week 6 lecture, censored data are those where _____

a. the value is missing for reasons unrelated to anything in the data, observed or not
b. we have partial knowledge of the actual value, such as knowing only that it is above a limit
c. values outside a range are not recorded at all, so those units are absent from the data
d. the same unit was measured more than once and the measurements disagree

20. From the Week 6 lecture, why is dropping incomplete cases a risk?

a. The cases that remain may differ from those dropped, which biases the sample.
b. Imputation is always more accurate than dropping cases, whatever the reason for the gap.
c. It is not allowed under the licence that governments attach to their data.
d. It always reduces the sample below the size needed for a confidence interval.

21. From the Week 6 lecture, the American Community Survey differs from the decennial census in that it _____

a. is run by a private polling company under contract to the Census Bureau
b. samples a share of households every year rather than counting everyone every ten years
c. counts every household every year rather than once every ten years
d. asks no questions about income, education, housing, or how people get to work

22. From the Week 6 lecture, in stratified sampling we _____

a. select every kth unit from a list of the population
b. give every unit an equal chance of selection and do nothing more than that
c. divide the population into groups and sample within every group
d. divide the population into groups and sample only some of the groups

23. From the Week 6 lecture, the difference between strata and clusters is _____

a. whether we intend to collect data from every group or only from some of them
b. whether the groups are geographic, such as states, or demographic, such as age groups
c. whether the groups overlap or are mutually exclusive
d. whether the groups were defined before or after the data were collected

24. From the Week 6 lecture, quota sampling is _____

a. sampling only the clusters that are cheapest to reach, then weighting them
b. sampling every kth unit from a list of the population
c. asking initial respondents to recruit further respondents from their networks
d. having strata but not sampling at random within them

25. From the Week 6 lecture, which of these is a probability sampling method?

a. Convenience sampling.
b. Quota sampling.
c. Snowball sampling.
d. Cluster sampling.

Logistic regression and cross-validation

26. From James et al. (2023), why not fit ordinary linear regression to a yes or no outcome?

a. Linear regression requires the predictors to be normally distributed.
b. Fitted values can fall below 0 or above 1, so they are not probabilities.
c. The coefficients of such a model cannot be tested.
d. Linear regression cannot be computed when the outcome is coded 0 and 1

27. From James et al. (2023), if the probability that a customer defaults is 0.2, the odds of default are _____

a. 0.8
b. 0.25
c. 0.2
d. 4

28. From James et al. (2023), a coefficient in a logistic regression measures _____

a. the change in the probability of the outcome for a one-unit change in the predictor
b. the odds of the outcome when the predictor is zero and the other predictors are at their means
c. the change in the log-odds of the outcome for a one-unit change in the predictor
d. the probability of the outcome when the predictor is at its mean

29. From James et al. (2023), a logistic regression of default on whether the customer is a student gives a coefficient of 0.40 on student, with predicted probabilities of default of 4.3 per cent for students and 2.9 per cent for non-students. The coefficient of 0.40 means that _____

a. the odds of default are 0.40 higher for students than they are for non-students
b. the predicted probability of default for a student is 0.40, and for a non-student it is the intercept
c. the log-odds of default are 0.40 higher for students, so their odds are about 1.5 times those of non-students
d. the probability of default is 0.40 higher for students than it is for non-students

30. From James et al. (2023), in logistic regression a one-unit change in X changes the probability of the outcome _____

a. by an amount that depends on the current value of X, because the curve is S-shaped
b. not at all, because only the log-odds change while the probability stays where it was
c. by the same percentage at every value of X, which is equal to the odds ratio
d. by the same amount at every value of X, equal to the coefficient

31. From James et al. (2023), how are the coefficients of a logistic regression estimated?

a. Maximum likelihood.
b. Cross-validation.
c. Ordinary least squares.
d. The method of moments.

32. From James et al. (2023), the outcome is a vote for Chow, Bradford, or someone else, so the appropriate model is _____

a. logistic regression, with the candidates coded as 1, 2, and 3
b. ordinal logistic regression, which handles three or more categories
c. linear regression, with the candidates coded as 1, 2, and 3
d. multinomial logistic regression, which handles unordered categories

33. From James et al. (2023), in the softmax coding of multinomial logistic regression, _____

a. each class gets a linear score, and a class's probability is its exponentiated score divided by the sum over all classes
b. one class is the baseline with a score of zero, and so its predicted probability is one
c. the class with the largest score is given probability one and the other classes are given probability zero
d. the probabilities differ from those under baseline coding, so the choice of coding changes the predictions

34. From James et al. (2023), why is the training error a poor guide to how well a model will predict?

a. It includes the irreducible error, which the test error leaves out, so it is always too high.
b. It keeps decreasing as the model becomes more flexible, even when the test error is increasing.
c. It is always larger than the test error, so it makes every model look worse than it is.
d. It cannot be computed for flexible methods, which have no residuals to sum.

35. From James et al. (2023), the main drawback of the validation set approach is that _____

a. the model has to be fitted once for every observation, which is slow
b. it always underestimates the test error, whichever observations are held out
c. the estimate depends on the split, and the model is fitted to only part of the data
d. it works for classification problems but not for regression problems

36. From James et al. (2023), in leave-one-out cross-validation _____

a. each predictor is left out once, and the model is fitted p times
b. a random tenth of the data is held out, and the model is fitted once
c. the data are split in half, and the model is fitted to each half
d. each observation is held out once, and the model is fitted n times

37. From James et al. (2023), compared with leave-one-out, 5-fold or 10-fold cross-validation has _____

a. the same estimate of the test error
b. slightly more bias but lower variance, and it is far cheaper to compute
c. less bias and less variance, which is why it is always preferred
d. more bias and more variance, so it is used only when time is short

38. From James et al. (2023), to choose between models of differing flexibility with cross-validation you _____

a. choose the model with the highest R^2 on the full dataset
b. choose the model with the most significant coefficients
c. choose the model with the lowest cross-validated error
d. choose the model with the lowest error on the training data

Part B: Short answer

B1. Describe multilevel regression and poststratification in your own words for the US election. Say what the cells are and where the population counts come from.

B2. Your team wants a sample of Toronto voters and can afford to survey 1,000 people. Compare stratified sampling by ward with cluster sampling by ward, and say which you would use and why.

B3. Explain why a quantile dotplot may serve a general reader better than the sentence “Chow has a 72 per cent chance”, and sketch one with 20 dots for that forecast.

B4. Two logistic regressions of whether a respondent would vote for Chow were fitted to the same respondents. Entries are coefficients on the log-odds scale.

Predictor	Model 1	Model 2
Intercept	-0.2	0.6
Renter	0.9	0.3
Age in years		-0.03

Interpret the renter coefficient in each model, and explain why it changed when age was added.

B5. Your team has two candidate models of vote choice, a logistic regression with four predictors and a much more flexible model. Describe how you would use 5-fold cross-validation to choose between them, and explain why comparing their errors on the training data would mislead you.

References

Gelman, Andrew, Jessica Hullman, Christopher Wlezien, and George Elliott Morris. 2020. "Information, Incentives, and Goals in Election Forecasts." *Judgment and Decision Making* 15 (5): 863–80.

James, Gareth, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor. 2023. *An Introduction to Statistical Learning with Applications in Python*. Springer. https://www.statlearning.com.

Kay, Matthew, Tara Kola, Jessica Hullman, and Sean Munson. 2016. "When (Ish) Is My Bus? User-Centered Visualizations of Uncertainty in Everyday, Mobile Predictive Systems." *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. https://doi.org/10.1145/2858036.2858558.

Wang, Wei, David Rothschild, Sharad Goel, and Andrew Gelman. 2015. "Forecasting Elections with Non-Representative Polls." *International Journal of Forecasting* 31 (3): 980–91. https://doi.org/10.1016/j.ijforecast.2014.06.001.